



Assignment of Basic Mathematics

Submitted by :-

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Semester - 1st

Submitted to :-

KUNDAN SIR

BCA-SEMESTER 1ST SYLLABUS

❖ **Symbolic Logic**

❖ **Set Theory**

❖ **Relation**

❖ **Function**

❖ **Differential Calculus**

- Successive differentiation
- Partial differentiation
- Maxima and Minima
- Tangent Normal
- Radius of carvature

❖ **Integral Calculas**

❖ **Indefinite Integration**

❖ **Defenite Integration its Properties**

❖ **Area , Lane and Volume if Curve**

❖ **Multiple Integral**

❖ **State line**

❖ **Pair of State line**

❖ **Circle**

❖ **Curve Tracing**

SYMBOLIC LOGIC

Propositional

A statement or proposition is a declarative sentence which is either true or false but not both.

Ex. 1. Delhi is the capital of india ;

2. Haryana is a country ;

3. 7 is a prime number ;;

4. $2+3=5$ 5. wish you a happy new year ;

6. how are you ;;

7. please wait

Truth value

If any proposition is true then its truth value is denoted by T and if the proposition is false then its truth value is denoted by F.

EX. 1 is less than 3 ----- T .

14 is odd number -----F.

Types of statements

1.Atomic Statement(Simple Sentences)

A statement which has no logical connective in it is called an Simple sentence.

Ex. a. This is my book b. Delhi is the capital of U.P.

2.Molecular or Compound Statement

The sentence formed by two simple sentence using a logical connective is called a compound sentence .

Ex. a. U.P is in India and Lucknow is the capital of U. P .

b. Hari will study or he will play.

Logical connectives

The particular words and symbols used to join two or more proposition into a single composite form or compound proposition are called logical connectives.

Logical connectives words	Symbol	USES
And/conjunction/join	$\wedge$	$p \wedge q$
Or/disjunction/meet	$\vee$	$p \vee q$
Negation	$\sim$	$p \sim q$
Equivalent	$\leftrightarrow$	$p \leftrightarrow q$
Conditional , if....then...	$\Rightarrow$	$p \Rightarrow q$
Biconditional , iff	$\Leftrightarrow$	$p \Leftrightarrow q$
NAND(NOT+AND)	$\uparrow$	$p \uparrow q$
NOR(NOT+OR)	$\downarrow$	$p \downarrow q$
XOR	$\oplus$	$p \oplus q$

Basic logical operation

1.conjunction:-

Any two proposition can be combined by the word “And” to form a compound proposition said to be the conjunction.

Truth table		
P	$p \wedge q$	$p \vee q$
T	T	T
T	F	F
F	T	F
F	F	F

EX. 1. Delhi is in India and $2+2=4$ -----T .

2. Delhi is in India and $2+2=5$ ----- F .

3. Delhi is in Russia and $2+2=4$ ----F.

2 . Disjunction :-

Any two proposition can be combined by the word “OR” to form a compound proposition said to be the Disjunction.

Truth table		
P	q	$p \wedge q$
T	T	T
T	F	T
F	T	T
F	F	F

EX. 1. Delhi is in India or $2+2=4$ ---- T .

2. Delhi is in India or $2+2=5$ ----- T.

3. Delhi is in Russia or $2+2=6$ -----F.

3 . Negation:-

if p is a statement then the negation of p is the statement not p denoted by $\sim p$.

Truth table	
p	$\sim p$
T	F
F	T

EX. p = Delhi is the capital of India .

$\sim p$ = Delhi is not the capital of India.

Conditional statement :-

Many statement are of the form “if p then q” such statement are said to be the conditional statement and denoted by $p \Rightarrow q$ or $p \rightarrow q$.

Truth table		
P	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

EX. If I am hungry then I will eat.

Biconditional statement :-

A statement “p if and only if q” such statement are said to be bi-conditional statement and denoted by $p \Leftrightarrow q$ or $p \leftrightarrow q$.

Truth table		
P	q	$P \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautologies:-

A proposition is said to be an tautologies if it contain only T in last column of truth table .

Contradiction:-

A proposition is said to be an tautologies if it contain only F in last column of truth table .

Contingency:-

A statement that can be either or false depending on the truth values of its propositional variables is called a contingency.

Converse of conditional statement:-

let $p \rightarrow q$ be a conditional statement then $q \rightarrow p$ is called its converse.

Ex. if X is a labourer then he is poor.

Its converse

If X is poor then he is labourer

Inverse of conditional statement:-

let $p \rightarrow q$ be a conditional statement then $\sim p \rightarrow \sim q$ is called its inverse.

Ex. if Anjesh read book then he will get knowledge ITS inverse.

If Anjesh not read book then he will not get knowledge.

Contrapositive of conditional statement:-

let $p \rightarrow q$ be a conditional statement then $\sim q \rightarrow \sim p$ is called contrapositive.

Ex. if Y is differentiable then it is continuous

Its contrapositive .

If Y is not continuous then it is not differentiable.

Logical Equivalence:-

The two statement p and q are said to be logically equivalent if they have the same truth values in all the possible cases.

Symbolically it is written as $p \equiv q$,

Symbolic logic question:-

❖ Truth table

Qn. $A \vee b \rightarrow b \wedge c$

Sol

a	b	c	$a \vee b$	$b \wedge c$	$a \vee b \rightarrow b \wedge c$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	F	F
F	T	T	T	T	T
F	F	T	F	F	T
F	T	F	T	F	F
T	F	F	T	F	F
F	F	F	F	F	T

Answer

SET THEORY

Sets:-

A collection of well define elements are caled set. Well define means common relation between collection of elements.

$$\text{Ex. } A = \{ 1, 2, 3, 4, 5, \dots \}$$

Some rule for writing in set

- ❖ A set is always denoted by capital letter and elements are denoted in small letter and elements are separate them by commas.
- ❖ A set is always written in mid bracket.
- ❖ Repetation is not allowed.
- ❖ Order are not necessary.

$$\text{Ex. } A = \{ 1, 2, 3 \}$$

$$A = \{ 1, 3, 2 \}$$

$$A = \{ 2, 3, 1 \}$$

$$A = \{ 2, 1, 3 \}$$

Some representation of set

1. **Roster method:** In this method we write collection of elements directly in the mid bracket and elements are separate them by commas.

$$\text{Example: } A = \{ 1, 2, 3, 4, 5, 6 \}$$

2. **Set builder method:** In this method we write common relation between collection of elements in the mid bracket.

Example: $A = \{ x : 1 < x < 6, x \in \mathbb{N} \}$

Note:- Every set can be represented in set builder method but every set cannot be represented in roster method.

Some types of set

- **Empty set:** Any set is called null set if it has no any element.
- Generally it is denoted by $\varnothing$ or $\{ \}$.

Example: $A = \{ x : 1 < x < 2, x \in \mathbb{N} \}$

$$B = \{ n : n = 2 + 3n \text{ and } n \in \mathbb{N} \}$$

$$B = \varnothing \text{ or } \{ \}$$

- **Singleton set:** Any set is called singleton if it has only one element.

Example: $A = \{ 3 \}$

- **Multi set:** Any set is called multi set if it has two or more elements.

Example: $A = \{ 1, 2, 3 \}$

- **Pair set:** Any set is called pair set if it has only two elements.

Example: $A = \{ 1, 2 \}$

- **Finite set:** Any set is called finite if it has finite number of elements.

Example: $A = \{ 2, 4, 6, \dots, 100 \}$

- **Infinite set:** Any set is called infinite set if it is not finite.

Example: $A = \{ 1, 2, 3, 4, \dots \}$

- **Non-empty set:** Any set if it has one or more elements then it is called non-empty set.

Example: $A = \{ 1, 2, 3 \}$

$B = \{ 2, 4, 6, 8, \dots \}$

- **Equal Set:** Let A and B be two sets is said to be equal $A \subseteq B$ and $B \subseteq A$ is called equal set.

Example: $A \subseteq B$ and $B \subseteq A$

$A = \{ 1, 2, 3 \}$ ans $\{ 3, 2, 1 \}$

- **Equivalent Set:-** Two sets A and B are said to be equivalent if the number of elements in set A is equal to the number of elements in set B.

Note:- Every equal set is an equivalent set, but the converse may or may not be true.

Types of Subset

- ❖ **Subset set:** Let A and B be two sets, then if each element of set B lies in set A then we say that B is a subset of A and A is a superset of B.

It is denoted by $B \subseteq A$ $A \supseteq B$

- 1) Total number of subsets equals to 2^n , where n = number of elements given set.
- 2) A finite set have minimum two subset.
1) Empty 2) Itself
- 3) A finite set have maximum 2^n subset.
- 4) Total number of non $\emptyset$ subset equals to 2^{n-1} .

Ex.

$$1. A = \{ 1 \}$$

$$\text{Ans} = \emptyset, \{ 1 \}$$

$$2. B = \{ 1, 2 \}$$

$$\text{Ans} = \emptyset, \{ 1 \}, \{ 2 \}, \{ 1, 2 \}$$

❖ **Proper Subset:** Let A and B be two sets, each elements of set B lies in set A but atleast one elements of set A does not lies in set B then we say that B is proper subset of A and A is called proper super set of B.

It is denoted by $B \subset A$;

Note:- Total number of proper subser equals to 2^{n-1} .

Ex.

$$1. A = \{ 1 \}$$

$$\text{Ans} = \emptyset$$

$$2. B = \{ 1, 2 \}$$

$$\text{Ans} = \emptyset, \{ 1 \}, \{ 2 \}$$

Operation on Sets

- ❖ **Union:-** Let A and B be two sets. Then the set of all those elements which lie in set A or set B or both A and B is called the Union of A and B.

It is denoted by $A \cup B$.

$$A \cup B = \{ x : x \in A \text{ or } x \in B \}$$

$$A = \{ 5, 6, 7, 8 \}, B = \{ 7, 8, 9, 10 \}$$

$$A \cup B = \{ 5, 6, 7, 8, 9, 10 \}$$

- ❖ **Intersection Set:-** Let A and B be two sets. Then the set of all those elements which lie in both A and B is called the Intersection of A and B.

It is denoted by $A \cap B$

$$A \cap B = \{ x : x \in A \text{ and } x \in B \}$$

$$A = \{ 5, 6, 7, 8 \} \text{ and } B = \{ 7, 8, 9, 10 \}$$

$$A \cap B = \{ 7, 8 \}$$

- ❖ **Difference of Sets:-** Let A and B be two sets. Then the set of all those elements of set A which do not lie in set B is called the difference of A and B. It is also referred to as the relative complement of B in A.

It is denoted by $A - B$.

$$A - B = \{ x : x \in A \text{ \& } x \notin B \}$$

$$A - B = A \cap B^c$$

$$A = \{ 1, 2, 3, 4 \}, B = \{ 3, 4, 5, 6 \}$$

$$A - B = \{ 1, 2 \}$$

$$B - A = \{ 5, 6 \}$$

❖ **Symmetric Difference:-** Let A and B be two sets. Then the set of all those elements which lie in set A or set B but not both A and B is called the Symmetric Difference of A and B.

Generally, it is denoted by $A \Delta B$.

$$A \Delta B = \{ x : x \in A \text{ or } x \in B \text{ \& } x \notin A \cap B \}$$

$$A \Delta B = (A - B) \cup (B - A)$$

$$A = \{ 1, 2, 3, 4 \}, B = \{ 3, 4, 5, 6 \}$$

$$A \Delta B = \{ 1, 2, 5, 6 \}$$

❖ **Universal Set:-** Any set is called a Universal Set if it is a super set of all the sets under consideration.

It is denoted by "U".

❖ **Complement Set:-** Let U be a Universal set and A be any set. Then $U - A$ is called the Complement of set A.

It is denoted by A^c or A' .

$$A^c = \{ x : x \in U \text{ \& } x \notin A \}$$

$$A = \{ 1, 2, 3 \}, B = \{ 1, 2, 3, 4, 5, 6, 7 \}$$

$$A^c = \{ 4, 5, 6, 7 \}$$

Sets formulas and questions

$$\diamond x \in A \cup B \Leftrightarrow x \in A \text{ or } x \in B$$

$$\diamond x \notin A \cup B \Leftrightarrow x \notin A \text{ and } x \notin B$$

$$\diamond x \in A \cap B \Leftrightarrow x \in A \text{ and } x \in B$$

$$\diamond x \notin A \cap B \Leftrightarrow x \notin A \text{ or } x \notin B$$

$$\diamond x \in A - B \Leftrightarrow x \in A \text{ and } x \notin B$$

$$\diamond x \in A \Leftrightarrow x \notin A^c$$

$$\diamond x \in A^c \Leftrightarrow x \notin A$$

Qn. $A \cup B = B \cup A$

Sol ~

$$= \text{Let } x \in A \cup B$$

$$= x \in A \text{ or } x \in B$$

$$= x \in B \text{ or } x \in A$$

$$= x \in B \cup A$$

$$= A \cup B \subseteq B \cup A \quad . \quad . \quad . \quad . \quad (1)$$

$$= x \in B \cup A$$

$$= x \in B \text{ or } x \in A$$

$$= x \in A \text{ or } x \in B$$

$$= x \in A \cup B$$

$$= B \cup A \subseteq A \cup B \quad . \quad . \quad . \quad . \quad (2)$$

$$= \text{From (1) to (2)}$$

$$= A \cup B = B \cup A$$

$$\text{LHS} = \text{RHS}$$

Proved

RELATION

Relation:-

Let A and B be two sets. A Relation R is said to be a subset of the Cartesian product $A \times B$.

Generally, it is denoted by R.

$$R = \{ (x,y) : xRy \}$$

• Example:-

If $A \times B = \{(1, 3), (2, 3)\}$, then possible relations are:

$$R_0 = \{ \}$$

$$R_1 = \{(1, 3)\}$$

$$R_2 = \{(2, 3)\}$$

$$R_3 = \{(1, 3), (2, 3)\}$$

Formulae for Number of Relations:

- If set A has m elements and set B has n elements, then the total number of relations from A to B is $2^{m \times n}$.
- If set A has m elements, then the total number of relations from A to A is 2^{m^2} .
- If set B has n elements, then the total number of relations from B to B is 2^{n^2} .

Types of Relations

❖ **Identity Relation:-** Every element of set A is related to only itself is called identity relation.

Generally, it is denoted by I_A .

$$I_A = \{(x,x) : \forall x \in A\}$$

• Example: If $A = \{ 1, 2, 3 \}$, then,

$$R = \{(1, 1), (2, 2), (3, 3)\}.$$

❖ **Reflexive Relation:-** A relation R defined on set A is said to be reflexive if xRx .

That is, $(x, x) \in R \forall x \in A$.

• Example: If $A = \{ 1, 2, 3 \}$, then

$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3)\}$ is reflexive.

❖ **Symmetric Relation:-** A relation R defined on set A is said to be symmetric if, $xRy \Leftrightarrow yRx$ for all $x, y \in A$.

That is, $(y, x) \in R \Leftrightarrow (x, y) \in R$ for $x, y \in A$

• Example: If $A = \{ 1, 2, 3 \}$, then

$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ is symmetric.

❖ **Anti-Symmetric Relation:-** A relation R defined on set A is said to be anti-symmetric if, $xRy, yRx \rightarrow x = y$ for $x, y \in A$.

That is, if $(x, y) \in R$ and $(y, x) \in R$, then $x = y$ for $x, y \in A$.

• Example: If $A = \{ 1, 2, 3 \}$, then,

$R = \{(1, 1), (2, 2), (1, 2)\}$ is anti-symmetric.

❖ **Transitive Relation:-** A relation R defined on set A is said to be transitive if, $xRy, yRz \Leftrightarrow xRz$ for all $x, y, z \in A$.

That is, $(x, y) \in R$ and $(y, z) \in R \Leftrightarrow (x, z) \in R$ for $x, y, z \in A$.

• Example: If $A = \{ 1, 2, 3 \}$, then,

$R = \{(1, 1), (2, 2), (1, 2), (2, 1)\}$ is not transitive unless it also contains $(1, 1)$ and $(2, 2)$.

❖ **Order Relation:-** A relation R defined on set A is said to be an order relation if, xRy or yRx for elements in the set.

• **Example:** If $A = \{ 1, 2, 3 \}$,

$$R = \{(1, 2), (2, 3), (1, 3)\}.$$

❖ **Equivalence Relation:-** A relation R define on set A is said to an equivalence relation if it is:

1. Reflexive
2. Symmetric
3. Transitive

❖ **Partial Order Relation:-** A relation R define on set A is said to be partial order relation if it is:

1. Reflexive
2. Anti-Symmetric
3. Transitive

❖ **Total Order Relation:-** A relation R define on set A is said to be total order relation if it is:

1. Reflexive
2. Anti-Symmetric
3. Transitive
4. Order Relation

FUNCTIONS

Functions:-

Let X and Y be two non-empty set each element of set x associate with unic elements of set Y under a fix rule those fix rule are called function.

Janerally It is denoted by $f : x \rightarrow Y$,

- **Domain:-** Those set were rule to be defined is called the Domain.

That is Set X.

- **Co-domain:-** The set of all possible output values is called the Co-domain.

That is Set Y.

- **Range:-** The set of all actual output values obtained from the function is called the Range

That is,

$$y = f(x)$$

Note:- Range is always a subset of Co-domain.

TYPES OF FUNCTIONS

1. One-One Function:- A function $f : X \rightarrow Y$ is called one-one function if different elements of domain have different images in codomain.

That is,

$$\text{If } f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

OR

$$\text{If } x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$$

2. Many-One Function:- A function is called many-one function if two or more elements of domain have the same image in codomain.

That is,

$$f(x_1) = f(x_2) \text{ where } x_1 \neq x_2$$

3. Onto Function:- A function $f : X \rightarrow Y$ is called onto function if every element of codomain is the image of at least one element of domain.

That is,

$$\text{Range} = \text{Co-domain}$$

4. Into Function:- A function is called into function if it is not onto.

That is,

$$\text{Range} \neq \text{Co-domain}$$

(Some elements of codomain are not images of any element of domain.)

5. One-One Onto Function:- A function is called one-one onto function if it is both one-one and onto.

That is,

- Different domain elements have different images
- Every element of codomain is an image

Composite Function:-

(i) $f \circ g$:- For $f \circ g$ to be defined.

- Range of $g \subseteq$ Domain of f
- New function maps: Domain of $g \rightarrow$ Co-domain of f

(ii) $g \circ f$:- For $g \circ f$ to be defined.

- Range of $f \subseteq$ Domain of g
- New function maps: Domain of $f \rightarrow$ Co-domain of g

EXAMPLES OF COMPOSITE FUNCTION

Example 1.

$$\text{Given: } f(x) = x^2 + 1$$

$$g(x) = \sqrt{x}$$

$$(i) (f \circ g)(x) = f(g(x))$$

$$= f(\sqrt{x})$$

$$= (\sqrt{x})^2 + 1$$

$$= x + 1$$

Answer

$$(ii) (g \circ f)(x) = g(f(x))$$

$$= g(x^2 + 1)$$

$$= \sqrt{(x^2 + 1)}$$

Answer

DIFFERENTIAL CALCULUS

Calculus:-

The branch of Mathematics in which study about change in variable is called calculus.

There are two types of variable :

- Independent variable ;
- Dependent variable ;
 - $y = f(x)$
 - $\frac{dy}{dx}$

There are two types of calculus:

- Differential calculus
- Integral calculus

Differential Calculus:- The branch of mathematics in which study about rate of change in variable is called differential calculus.

It is denoted by:

$$\frac{dy}{dx}, y'(x), f'(x), f_x(x)$$

Some basic formula for differentiation

$$1. \frac{dy}{dx} (x^n) = n x^{n-1}$$

$$2. \frac{dy}{dx} (e^x) = e^x$$

$$3. \frac{dy}{dx} (\log x) = \frac{1}{x}$$

$$4. \frac{dy}{dx} (\sin x) = \cos x$$

$$5. \frac{dy}{dx} (\cos x) = -\sin x$$

$$6. \frac{dy}{dx} (\tan x) = \sec^2 x$$

$$7. \frac{dy}{dx} (\cot x) = -\operatorname{cosec}^2 x$$

$$8. \frac{dy}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$$

$$9. \frac{dy}{dx} (k) = 0$$

$$10. \frac{dy}{dx} (a^x) = a^x \log a$$

$$11. \frac{dy}{dx} (\sqrt{x}) = \frac{1}{(2\sqrt{x})}$$

$$12. \frac{dy}{dx} (x) = 1$$

$$13. \frac{dy}{dx} \left(\frac{1}{x}\right) = \frac{-1}{x^2}$$

$$14. \frac{dy}{dx} (k f(x)) = k \frac{d}{dx} (f(x))$$

• U.V method

➤ $y = u.v$

➤ $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

• $\frac{U}{V}$ method

➤ $Y = \frac{U}{V}$

➤ $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} / V^2$ $(V \neq 0)$

SUCCESSIVE DIFFERENTIATION:-

Successive differentiation:-

Differentiate any function more than one time continuously is called successive differentiation.

Generally it is denoted by ;

$$y_n \text{ or } y^n$$

$y = e^{ax}$ then find y_n .

$$y_1 = e^{ax} \times a$$

$$y_2 = e^{ax} \times a$$

$$y_3 = e^{ax} \times a$$

$$y_n = e^x \times a^n$$

PARTIAL DIFFERENTIATION

Partial derivative:-

A partial derivative is the derivative at multi variable function with respect to one at its variable on another variable treat like constant.

It is denoted by, $\frac{d}{dx}, \frac{d}{dy}, \frac{d}{dz}$.

Qn. $U = x^2 + y^2 + xy$

Sol $\frac{\partial U}{\partial y} = 2x + y^2 + xy$

$$= 2x + y$$

$$\frac{\partial U}{\partial y} = \frac{0 + 2y + x}{2y + x}$$

$$\frac{\partial^2 U}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial U}{\partial x} \right)$$

$$= \frac{\partial}{\partial x} (2x + y)$$

Answer

MAXIMA AND MINIMA

Maxima:-

Let $f(x)$ be the function of one variable defined on domain D .

Then function is maximum,

If, $f(x) \leq f(a)$

Minima:-

Let $f(x)$ be the function of one variable defined on domain D .

Then function is minimum,

If, $f(x) > f(a)$

Stationary Point / Critical Point

Those point of domain where tangent are parallel to domain are called stationary point.

Extreme Point

Those point of domain where function take extreme (maxm, minm) value are extreme point.

Note :-

- i) Every extreme point is stationary point but every stationary point may or may not be true extreme point.
- ii) Maximum and minimum is unique but maxm and minm point are not unique.

Rule for finding maxm and minm (second derivative test)

Let $f(x)$ be the function of one (1) variable. firstly find the $\frac{dy}{dx}$, equating to 0 and solve for x that is,

i) $F(x)$

a) $f'(x)$

b) $f'(x) = 0$

c) $x = \alpha_1, \alpha_2, \alpha_3 \dots \alpha_n,$

ii) $F''(x)$

a) $F''(\alpha) < 0$

$X = \alpha$ is $\max^m$ point,

$F(\alpha)$ is $\max^m$ value,

b) $F''(\alpha) > 0$

$X = \alpha$ is $\min^m$ point

$F(\alpha)$ is $\min^m$ value

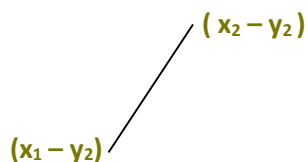
c) $F''(\alpha) = 0$

$X = \alpha$ is neither maximum nor minimum

TANGENT AND NORMAL

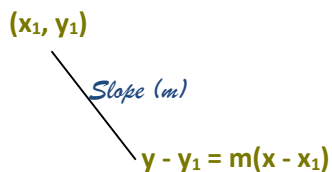
Line Equation Form:-

(i) Two Point Form.

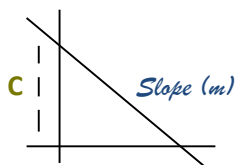


$$y - y_1 = \frac{y_2 - y_1}{x_1 - y_1}$$

(ii) Slope Point Form.

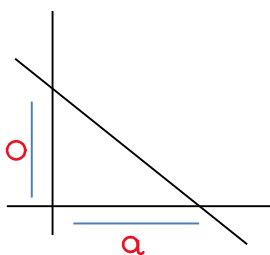


(iii) Slope Intercept Form.



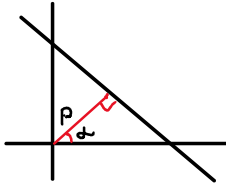
$$y = mx + c$$

(iv) Intercept Form.



$$\frac{x}{a} + \frac{y}{b} = 1$$

(v) Normal Form.



$$x \cos \alpha + y \sin \alpha = p$$

Tangent:-

Let P and Q be any two point on curve $y = f(x)$. when the point Q moving along the curve and coincide with point P the limiting of chord PQ becomes a Tangent at point P.

Slope of Tangent:-

i) If two point are given then slope of tangent.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

ii) If angle are given then slope of tangent.

$$m = \tan \theta$$

iii) If function are given then slope of tangent.

$$m = \left(\frac{dy}{dx} \right)_{(x, y)}$$

Slope of Normal

$$\text{Slope of Normal} = - \frac{1}{\text{Slope of Tangent}}$$

$$m_1 m_2 = -1$$

Equation of Tangent and Normal

$$\text{Equation of Tangent} = y - y_1 = m(x - x_1)$$

$$\text{Equation of Normal} = y - y_1 = -\frac{1}{m}(x - x_1)$$

$$\text{Length of Sub Normal} = y \cdot \frac{dy}{dx}$$

$$\text{Length of Normal} = y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

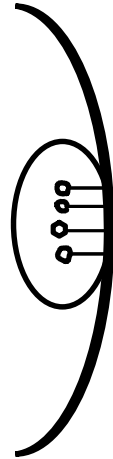
$$\text{Length of Sub Tangent} = \frac{y}{\frac{dy}{dx}}$$

$$\text{Length of Tangent} = \frac{y}{\frac{dy}{dx}} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

RADIUS OF CURVATURE

Radius of Curvature

Let P be any point on the curve. Q and R be any other two points on the curve which are very close to P. It is known that one and only one circle can pass through three points P, Q, R. When Q and R tend to P and coincide with P, then the limiting condition of the circle is called the Radius of Curvature.



- Radius of Curvature is denoted by $\rho(x)$.
- Its centre is called centre of curvature.
- Its radius is called radius of curvature.
- Its diameter is called diameter of curvature.
- Its circumference is called circumference of curvature.

Formula:-

$$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}}$$

INTEGRAL CALCULAS

There are two types of Integration

i) Indefinite integration

$$\int f(x) dx = f(x) + c$$

(Integral sign) *(Variable of Integration)* *(Orbitary constant)* *(Integrate)* *Integral Value*

ii) Difinite integration

$$\int_a^{\beta} f(x) dx = F(\beta) - F(a)$$

(Integral sign) *(upper limit)* *(Integrate)* *(lower limit)* *Integral Value*

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + c \quad \{ \because n \neq -1 \}$$

$$\begin{aligned} \text{Sol} &= \int x^2 \\ &= \frac{x^{2+1}}{2+1} + c \\ &= \frac{x^3}{3} + c \end{aligned}$$

Answer

Linear

$$1. \int (ax \pm b)^n dx = \frac{(ax \pm b)^{n+1}}{a(n+1)} + c$$

$$\begin{aligned} 2. \int (2x+3)^4 dx &= \frac{(2x+3)^{4+1}}{2(4+1)} + c \\ &= \frac{(2x+3)^5}{(10)} + c \end{aligned}$$

$$3. \int x^{-3} dx = \frac{x^{-3+1}}{-3+1} + c = \frac{1}{2} x^{-2} + c$$

$$4. \int x^{-1} dx = \int \frac{1}{x} dx = \log |x| + c$$

$$5. \int \frac{dx}{ax \pm b} = \frac{\log |ax \pm b|}{a} + c$$

$$6. \int \frac{dx}{5x+4} = \frac{\log |5x+4|}{5} + c$$

$$7. \int e^x dx = e^x + c$$

$$8. \int \sin x dx = -\cos x + C$$

$$9. \int \cos x dx = \sin x + c$$

$$10. \int \sec^2 x dx = \tan x + C$$

$$11. \int \sec x dx = \log |\sec x + \tan x| + C$$

$$12. \int \operatorname{cosec} x dx = \log |\operatorname{cosec} x - \cot x| + C$$

$$13. \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$14. \int \sec x \tan x dx = \sec x + C$$

$$15. \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$16. \int \tan x dx = \log |\sec x| + C$$

$$17. \int \cot x dx = \log |\sin x| + C$$

INTEGRATION BY PARTS

$$\int u \cdot v \, dx = u \int v \, dx - \int \left(\frac{du}{dx} \cdot \int v \, dx \right) dx$$

I – Inverse ($\sin^{-x}$, $\cos^{-x}$)

L – Lagarithmic ($\log x$)

A – Algebra (x , y , x^2 , y^2)

T – Trigonometric ($\sin x$, $\cos x$)

E – Exponential (e^x)

Qn. $\int x \cdot \sin x \, dx$

$$\begin{aligned} \text{Sol}^n &= u \int v \, dx - \int \left(\frac{du}{dx} \cdot \int v \, dx \right) dx \\ &= x \int \sin x \, dx - \int \left(\frac{dx}{dx} \cdot \int \sin x \, dx \right) dx \\ &= -x \cos x - \int 1 \cos x \, dx \\ &= -x \cos x - (-\sin x) \\ &= \boxed{-x \cos x + \sin x + c} \end{aligned}$$

Answer

TRANSFORMATION METHOD

$$a^2 - x^2 = \text{put } x = a \sin \theta / a \cos \theta$$

$$a^2 - x^2 / x^2 - a^2 = \text{put } x = a \tan \theta / a \cot \theta$$

$$x^2 - a^2 = \text{put } x = a \sec \theta / a \operatorname{cosec} \theta$$

$$\sqrt{a-x} / \sqrt{\frac{a-x}{a+x}} / \sqrt{a+x} / \sqrt{\frac{a+x}{a-x}} = \text{put } x = a \cos 2\theta$$

$$\text{Note:- i) } \sin 2x = \sin x \cos x$$

$$\text{ii) } \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$\text{iii) } \cot^2 x = \frac{1 + \cos 2x}{2}$$

$$\text{Qn. } \int \frac{dx}{1+x^2} = ?$$

Sol

$$\Rightarrow x = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1} x$$

$$\Rightarrow x = \tan \theta$$

$$\Rightarrow dx = \sec^2 \theta$$

$$\Rightarrow \int \frac{\sec^2 x \, d\theta}{1 + \tan^2 \theta}$$

$$\Rightarrow \int \frac{\sec^2 \theta}{\sec^2 \theta} d\theta$$

$$\Rightarrow \int d\theta$$

$$\Rightarrow \boxed{\theta + c}$$

Answer

PROPERTIES OF DEFINE INTEGRATION

1. $\int_a^b f(x) dx = ?$

$$\text{Sol} = \int_1^2 x^2 dx$$

$$= \left[\frac{x^3}{3} \right]_1^2$$

$$= \frac{8}{3} - \frac{1}{3}$$

$$= \frac{7}{3} \quad \text{Answer}$$

Put $x = z$

$dx = dz$

$x = 1$ then,

$z = 1$

it, $x = 2$

$z = 2$

2. $\int_a^b f(x) dx = - \int_a^b f(x) dx$

3. $\int_a^b f(x) dx = \int_b^e f(x) dx + \int_a^b f(x) dx$

$$[\sin x = - \sin x]$$

$$= \int_a^c f(x) dx + \int_c^d f(x) dx + \int_d^b f(x) dx$$

$$4. \int_{-a}^a f(x) dx = 0 \text{ If } f(x) \text{ is odd}$$

$$2 \int_0^a f(x) dx, \text{ If } f(x) \text{ is even}$$

$$5. \int_0^a f(x) dx = \int_b^a f(a-x) dx$$

$$6. \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

STRAIGHT LINE

Straight line

A line is of the form $ax + by + c = 0$.

A straight line is a curve such that every point on the curve lies between any two points on it.

General equation of straight line

$$Ax + By + c = 0$$

Equation of straight line

$$1) \text{ Slope Intercept form } = y = mx + c$$

$$2) \text{ Point Slope form } = y - y_1 = m(x - x_1)$$

$$3. \text{ Two-Point form } = y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$4) \text{ Intercept form } = \frac{x}{a} + \frac{y}{b} = 1$$

Distance from a point (x_1, y_1) to a line. $ax + by + c = 0$

$$\text{Distance (d)} = \left| \frac{Ax + By + C}{\sqrt{A^2 + B^2}} \right|$$

Angle between two lines with slopes, m_1 & m_2

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$
$$\theta = \tan^{-1} \left(\frac{m_2 - m_1}{1 + m_1 m_2} \right)$$

Equation of straight line

i) If 2 lines are perpendicular then

$$m_1 m_2 = -1$$

ii) If 2 lines are parallel then,

$$m_1 = m_2$$

PAIR OF STRAIGHT LINES

Pair of straight lines

A second degree equation is pair of straight line, this invoice second equation represent sraight line usually in the form of a second hemogineous equation.

The equation: $ax^2 + 2hxy + by^2 = 0$ always represents 2nd pair of straight lines passing through the origin.

1. Angle between them :

$$\tan \theta = 2 \sqrt{\frac{h^2 - ab}{a+b}}$$

2. Condition for perpendicular lines :

$$a + b = 0$$

3. Condition for parallel lines:

$$h^2 - ab = 0$$

4. General second degree equation :

$$ax^2 + 2hxy + by^2 + 2yx + 2hy + c = 0$$

REPRESENT A PAIR OF STRAIGHT LINES

Qn. $3x^2 + 2xy - y^2 = 0$

Sol

$$ax^2 + 2hxy + by^2$$

$$a = 3, 2h = 2, h = -1$$

$$a = 3, 2h = 2, b = -1$$

$$m^2 - 2m - 3 = 0$$

$$m^2 - 2m + m - 3 = 0$$

$$\text{Put, } y = mx$$

$$3x^2 + 2xmx - m^2 x^2 = 0$$

$$x^2(3 + 2m - m^2) = 0$$

$$-x^2(m^2 - 2m - 3) = 0$$

$$y = mx$$

$$y = 3x$$

$$y = -x$$

$$(y - 3x)(y + x)$$

$$y^2 - 3xy + xy - 3x^2$$

$$y^2 - 2xy - 3x^2 = 0$$

$$3x^2 + 2xy - y^2 = 0$$

Answer

CIRCLE

Circle

A circle is the locus of a point which moves in a plane such that its distance from a fixed point is always constant.

- **General equation of circle :**

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

- **Centre (c) is (-g, -f) :**

$$\text{Radius (r) is } \sqrt{g^2 + f^2 - c}$$

- **Standard equation of circle :**

If centre is origin, then equation of circle is:

$$x^2 + y^2 = r^2$$

If centre is (h, k),

Then equation of circle is:

$$(x - h)^2 + (y - k)^2 = r^2$$

CURVE TRACING

Symmetric (Mirror Image):-

About x-axis:- Let $f(x, y) = 0$ be the given equation of curve. replace y by $-y$ in the given equation of curve, if the given equation remains unchanged, then the curve is symmetric about x-axis.

That is,

$$f(x, y) = f(x, -y)$$

If degree of each y is even in the given equation of curve, then curve is symmetric about x-axis.

About y-axis:- Let $f(x, y) = 0$ be the given equation of curve, replace x by $-x$ in the given equation of curve. If the given equation remains unchanged, then curve is symmetric about y-axis.

That is,

$$f(x, y) = f(-x, y)$$

If degree of each x is even in the given equation of curve, then curve is symmetric about y-axis.

About Origin:- Let $f(x, y) = 0$ be the given equation of curve. Replace x by $-x$ and y by $-y$ in the given equation of

curve. If equation remains unchanged, then curve is symmetric about origin.

That is,

$$f(x,y) = f(-x,-y)$$

About $y = x$ (Any line):- Let $f(x,y)=0$ be the given equation of curve.

Replace x by y and y by x in the given equation of curve. If equation remains unchanged, the curve is symmetric about the line $y = x$.

That is,

$$f(x,y) = f(y,x)$$

Intersection

i) With x-axis:- Let $f(x,y)=0$ be the given equation of curve. Put $y = 0$ in the given equation and solve for x . Then the real value of x is the intersection of curve with x -axis.

ii) With y-axis:- Let $f(x,y)=0$ be the given equation of curve. Put $x = 0$ in the given equation and solve for y . Then the real value of y is the intersection of curve with y -axis.

iii) **With $y = x$:-** Let $f(x,y)=0$ be the given equation of curve. Put $y = x$ in the given equation and solve for x . Then the real value of x is the intersection of curve with the line $y = x$.

1. Area of Plane Curve

The area bounded by the curve $y = f(x)$, x -axis, at $x = a$ and $x = b$ is given by,

$$\text{Area} = \int_a^b y \cdot dx$$

2.

The area bounded by the curve $x = f(y)$, y -axis, and $y = a$ to $y = b$ is given by,

$$\text{Area} = \int_a^b x \cdot dy$$

3. Length of Plane Curve

Let $y = f(x)$ be the equation of curve. Then its length or perimeter from $x = a$ to $x = b$ is given by,

$$\text{Length} = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot dx$$

4. Volume of Solid

Let $y = f(x)$ be the curve bounded by $x = a$ to $x = b$. Then the volume of solid is given by,

$$V = \pi \int_a^b y^2 \cdot dx$$

5. Surface of Solid

Let $y = f(x)$ be the curve bounded by $x = a$ to $x = b$. The surface area of solid is given by,

$$S = 2\pi \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \cdot dx$$